

6th Analytical Solved S.Q's

(2018)

(i)

Doppler broadening:

Doppler broadening is a consequence of phenomena similar to the familiar Doppler effect for sound, which results in change in pitch or frequency as a sound emitting source approaches and then passes an observer, for example for a passing train that is blowing its whistle.

Pressure broadening:

Pressure broadening (also called collision broadening) is brought about by collision between molecules or atoms, which can supply or remove small amounts of energy during radiative transitions, thereby allowing photons with a broader range of frequencies to produce a particular transition of a molecule.

(ii)

Characteristics of weak base anion resins:

WBA resins have amin functional groups, and are typically regenerated with sodium hydroxide (NaOH), ammonia (NH_3) or sodium carbonate (Na_2CO_3).

The weak base anion exchange resin is easier to regenerate than the strong base resin and can be efficiently regenerated with a small amount of sodium hydroxide.

Advantages of weak base anion resins:

WBA resins are commonly used in pharmaceutical, food engineering and chemical engineering. WBA resins are very useful for the removal of mineral acid, organic acids and other organic materials, especially for removal of high molecular weight organic matters from aqueous solutions under low pH condition.

(iii)

Samples in capillary electrophoresis system:

In capillary electrophoresis (CE), the conducting buffer is retained within a capillary tube whose inner diameter is typically 25-75 μm . Samples are introduced into one end of the capillary tube. As the sample migrates through the capillary its components separate and elute from the column at different times. A typical capillary electrophoresis system consists of a high-voltage power supply, a sample introduction system, a capillary tube, a detector and an output device. Some instruments include a temperature control device to ensure reproducible results.

(iv)

Percent extraction:

The fraction of solute extracted is equal to the millimoles of solute in the organic layer divided by the total number of millimoles of solute. The millimoles are given by the molarity times the milliliters. Thus, the percent extracted it is given by

$$\% E = \frac{[S]_o V_o}{[S]_o V_o + [S]_a V_a} \times 100\%$$

(v)

Electro-osmotic flow:

When an electric field is applied to a capillary filled with an aqueous buffer we expect the buffer's ions to migrate in response to their electrophoretic mobility. Because the solvent, H_2O , is neutral we might reasonably expect it to remain stationary. Under normal conditions we observed that the buffer moves towards the cathode. This phenomenon is called the electro-osmotic flow. Electro-osmotic flow occurs because the walls of the capillary tubing carry a charge.

(vi)

Role of masking in solvent extractions:

These agents, which are metal complexing agents, prevent particular metals from taking part in their usual reactions and in this way they remove their interference without requiring an actual separation. These agents also known as sequestering agents, are used to prevent certain metals from forming extractable complexes in solvent extraction and thus increases the selectivity of the extraction methods.

Role of oxidation state :

The selectivity of extraction may sometimes be increased by modifying the oxidation state of metal or the impure. Variation of the valance states of some elements in the organic extract is also useful as a method of stripping.

(vii)

Air-acetylene flame:

The air-acetylene flame is the most widely used flame. It is stable, simple to operate and produces sufficient atomization to enable good sensitivity and

Freedom from interferences for many elements. It is not only necessary for the flame to atomize the analyte, but also to break down and refractory compounds which may react with or physically entrap the analyte.

Nitrous oxide-acetylene flames:

The nitrous oxide-acetylene flame is both hot and reducing. A characteristic red, interconal zone is obtained under slightly fuel-rich conditions. This red feather is due to emission by the cyanogen radical. This radical is a very efficient scavenger for oxygen.

(viii)

Chemical interference in atomic spectroscopy:

If a sample contains a species which forms a thermally stable compound with the analyte that is not completely decomposed by the energy available in the flame then chemical interference exists. Refractory elements (Ti, W, Zr, Mo & Al) may combine with oxygen to form thermally stable oxides. Analysis of such elements can be carried out at higher flame temperatures using nitrous oxides-acetylene flame instead of air-acetylene to provide higher dissociation energy. Alternately an excess of another element or compound can be added e.g. Ca in presence of phosphate produces stable calcium phosphate which reduces absorption due to Ca ion.

(ix)

Principle of solid Phase extraction:

Solid phase extraction (SPE) is a reference sample preparation technique in modern laboratories. SPE permits to process samples from different nature, providing an improvement of the sensitivity and selectivity of the analytical determinations. The basic principles of SPE are similar to liquid-liquid extraction.

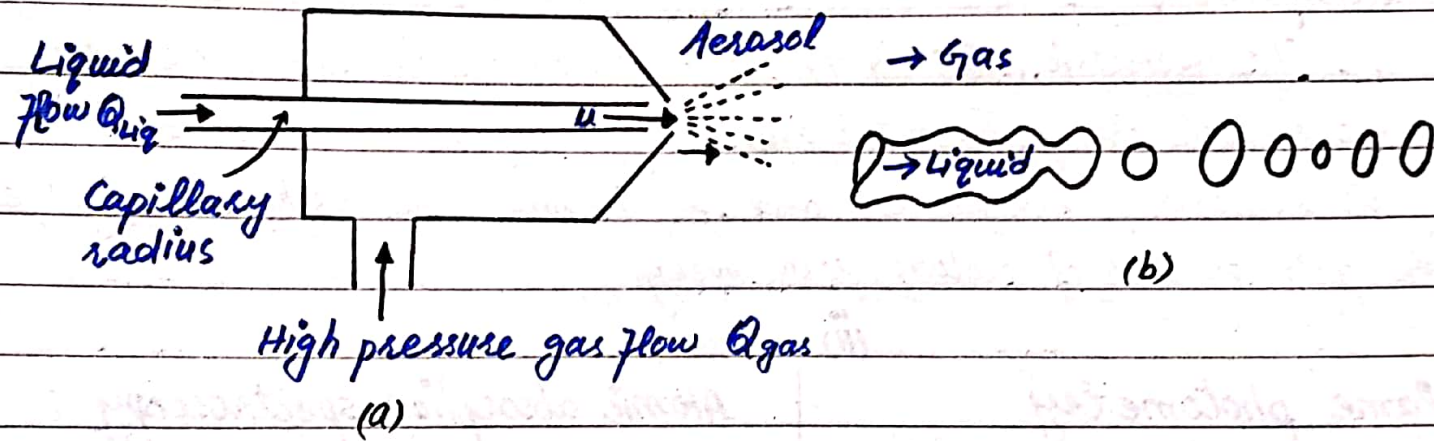
Both methods involve the distribution of dissolved species between two phases. However, SPE involves the dispersion between analyte between a liquid (sample medium) and a solid (adsorbent) phase instead of two liquid phases which cannot be mixed together as in liquid-liquid extraction. This technique allows the enrichment and purification of the analytes on a solid adsorbent through adsorption from the solution.

(x)

Nebulization:

Nebulization is a process of conversion of sample to a fine mist of finely divided droplets using a jet of compressed gas. Sample solution is introduced through an orifice into a high velocity gas jets, usually the oxidant, in either parallel

or perpendicular manner.



Sample stream is converted into a cloud of droplet in the aerosol modifier or spray chamber, combined with the oxidizer/fuel and carried to the burner.

(2019)

(i)

Isoelectric point:

The isoelectric point is the pH at which a particular molecule carries no net electric charge. The net charge on the molecule is affected by the pH of its surrounding environment and can become more positive or negative due to the gain or loss of protons, respectively.

(ii)

Flame photometry

A photoelectric flame photometry is a branch of atomic spectroscopy is used for inorganic chemical analysis for determining the concentration of certain metal ions such as sodium, potassium, lithium, calcium etc.

In flame photometry the species (metal ions) used in the spectrum are in the form of atoms.

Atomic absorption spectroscopy

Atomic absorption spectroscopy is an analytical technique used to determine how much of certain elements are in a sample. It uses the principle that atoms & ions can absorb light at a specific, unique wavelength.

When this specific wavelength of light is provided the energy (light) is absorbed by the atom.

(iii)

Examples of strong acid cationic resins:

Zerolite ZCS, Amberlite 120, Dowex 50, Sodium polystyrene sulfonate and sodium zirconium cyclosilicate are examples of strong acid cationic exchange resin.

(iv)

Percent extraction: Repeated 20/18 (iv)

(v)

Principle of AAS:

This technique uses basically the principle that free atoms (gas) generated in an atomizer can absorb radiation at specific frequency. AAS quantifies the absorption of ground state atoms in gaseous state. The atoms absorb UV or visible light and make transitions to higher electronic energy levels. The analyte concentration is determined from the amount of absorption. Atomic absorption is a very common technique for detecting metals and metalloids in environmental samples.

(vi)

Significance of Gel chromatography:

This method is especially useful for separating enzymes, proteins, peptides and amino acids from each other and from substances of low molecular weight. The separation of the components of a mixture by gel chromatography is based on the differences in the molecular size of components. In gel chromatography separation can be performed under conditions specifically designed to maintain the stability & activity of molecules of interest without compromising resolution.

(vii)

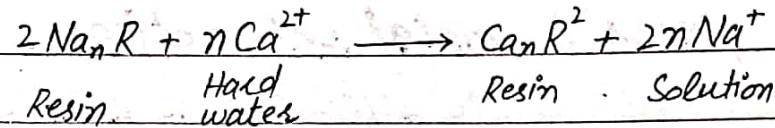
Electro-osmotic flow: Repeated 2018(v)

(viii)

Role of ion exchange chromatography in softening of hard waters

We are aware of the fact that the hardness of water is due to the presence of Ca^{2+} , Mg^{2+} and other divalent ions. These ions may be removed by passing hard water through cation exchangers charged with Na^+ when the following exchange

reaction takes place.



The Ca^{2+} and Mg^{2+} ions are retained in the column whereas the Na^+ ions pass into the solution. These Na^+ ions are harmless for washing purposes.

(ix)

Solid phase extraction:

Solid phase extraction (SPE) involves the separation of components of samples in solution through their selective interaction with the retention by a solid, particulates sorbent. SPE depends on differences in the affinities of the various components of the sample for the sorbent. The mechanisms of the interactions are virtually identical to the sorption processes that forms the basis of liquid chromatographic separations. SPE is a reference sample preparation technique in modern laboratories. SPE permits to process samples from different nature, providing an improvement of the sensitivity and selectivity of the determinations.

(x)

Role of complexing agents in metal extraction:

Many metals can be separated on anion exchange columns by being converted to anions by complexation. The complexing agent is an anion such as chloride, bromide or fluoride. Uncharged complexing agents also affect the equilibrium if they form a complex that must dissociate before the metal is exchanged or if they change its size. Many complexing agents are either weak acids or bases or are salts of these, and so complicated interdependence of pH and complexation often results.

~~— (2020) —~~

(i)

Sample in capillary electrophoresis system:

Repeated 2018 (iii)

(ii)

Efficiency of solvent extractions:

Repeated 2018 (iv), 2019 (iv)

(iii)

Electrophoretic mobility:

The velocity with which a solute moves in response to the applied electric field is called its electrophoretic velocity, v_{ep} ; it defined as

$$v_{ep} = \mu_{ep} E$$

Where μ_{ep} is the solute's electrophoretic mobility, which is defined as

$$\mu_{ep} = q / 6\pi\eta r$$

Where q is the solute's charge, η is the buffer's viscosity and r is the solute's radius.

Electrophoretic mobility is the ratio of the velocity of the ion to field strength. It is

14
directly proportional to net charge and inversely proportional to molecular size & viscosity of the electrophoretic medium.

Unit:

Its typical units are $\mu\text{m}\cdot\text{cm}/\text{V}\cdot\text{s}$ (micrometer centimeter per Volt second) since it is a velocity $[\mu\text{m}/\text{s}]$ per field strength $[\text{V}/\text{cm}]$

(iv)

Role of complexing agent in metal extractions:

Repeated 2019 (x)

(v)

Nernst Distribution Law:

This law can be stated as:

At constant temperature, when two solutions of a solute in two immiscible solvents are in contact with each other, then solute distributes itself to keep a fixed ratio of concentration in the two immiscible solvent at equilibrium.

Examples

The most useful example of Nernst distribution law is the extraction of a material dissolved in one solvent by another solvent, the two solvents being partially miscible or immiscible.

(vi)

AAS limited to metals:

Atomic absorption spectroscopy is limited to metals only. A major reason for this is that the atoms in metal elements are more easily readable. For AAS to be effective, the atoms in a material must be in isolation and free of possible contaminating lines from molecules. Metals generally have narrow, single emission and absorption lines, which form brightly and clearly.

(vii)

Gel chromatography known as size exclusion chromatography:

Gel chromatography also known as size exclusion chromatography because size exclusion chromatography (SEC) separates molecules based on their size by filtration through a gel. The gel consist of spherical beads containing pores of

a specific size distribution. Separation occurs when molecules of different sizes are included or excluded from the pores within the matrix. Gel chromatography is also known as size exclusion chromatography is used to separating different size of molecules.

(viii)

Ion exchangers:

Ion exchangers are natural substances for example certain clays or deliberately synthesized resins containing positive ions (cation exchangers) or negative ions anion (anion exchangers) that exchange with those ions in solution having a greater affinity for the exchanger.

Types:

i-Cation exchangers: There are further two types of cation exchange resins.

(a) Strong acid cation exchanger resin:

Examples: Zeolite ZCS, Amberlite 120, Dowex 50

(b) Weak acid cation exchanger resin:

Examples: Zeolite 226, Amberlite 50, Duolite C-63

ii- **Anion exchangers:** These are further two types of anion exchange resins.

(a) **Strong base anion exchanger resin:**

Examples: Zeolite FF, Amberlite 400, Dowex A-1

(b) **Weak base anion exchanger resin:**

Examples: Zeolite H, Amberlite -45, Duolite A-1 to 7.

(ix)

Limitations of Flame photometry:

Flame photometry has following limitations or disadvantages:

- 1- The concentration of the metal ion in the solution cannot be measured accurately.
- 2- It is difficult to obtain the accurate results of ions with higher concentration.
- 3- The information about the molecular structure of the compound present in this sample solution cannot be determined.
- 4- The elements such as carbon, hydrogen and halides cannot be detected due to its non radiating nature.

(X)

Atomic absorption spectroscopy

- This technique is superior & specific because of the fact that only the atoms of particular element can absorb radiations of their own characteristic wavelength.

- A relatively large number of metal atoms produce an atomic absorption signal whereby the effect of flame-temperature variation is negligible in AAS, i.e., independent of flame-temperature.

- The detection limits of sensitivity of the following elements are more by AAS technique such as: Ag, As, Au, B, Bi, Cd, Co & Fe.

Flame emission spectroscopy

- Spectral interferences usually take place in this technique.

- A much smaller number of metal atoms do produce an emission signal in FES, showing that this technique is not independent of flame-temperature.

- The detection limits of sensitivity of the following elements are more by FES technique such as: Al, Ba, Ca, Eu, Ho, In, K and La.

Note: This question is same as 2019(m)

—(2021)—

(i)

Compare air-acetylene & Nitrous oxide-acetylene flames in AAS:

Repeated 2018 (vii)

(ii)

Chemical interference: Repeated 2018 (viii)

(iii)

Principle of SPE: Repeated 2018 (ix)

(iv)

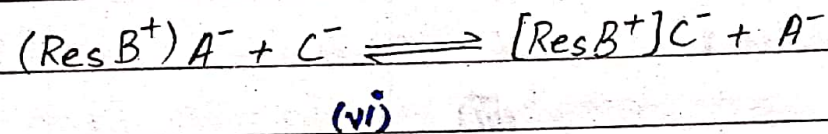
Percent extraction: Repeated 2018 (iv), 2019 (iv)

(v)

Characteristics of anion exchange resin:

Anion exchange resin is a high molecular weight cross linked polymer having $-N^+(R)_3$, $-N^+(R)_2$, $-N^+H_2(R)$, $-N^+H_3$ groups as an integral part of the resins, and equivalent amount of anion such as Cl^- , SO_4^{2-} , OH^- , etc. Thus an anion exchange resins contain a polymeric cation to which active

anions are attached. Thus an anion exchange resin can be represented as $(\text{Res B}^+) \text{A}^-$, where Res is the basic polymer resin to which the cation B^+ and a mobile anion A^- are attached. Where an anion exchange resin is placed in the solution, the mobile or active anions of the resin are exchanged for anions of the solution. Thus



Electro-osmotic flow: Repeated 2018 (v), 2019 (vii)
(vii)

Pack column with dry gel in Gel chromatography:

The slurry or gel inside the column should never be allowed to dry out. If this happens it may create cracks and unevenness in the solid phase, which will decrease the efficiency of the separation. Dry packing of the column followed by the liquid will result in cracking of column due to swelling of the gels. For these reasons dry gel is not suitable to pack the column in gel chromatography.

(viii)

Role of ion exchange chromatography in softening of hard water:

Repeated 2019 (viii)

(ix)

Distribution coefficient

The ratio of the equilibrium concentrations in the two phases defines as distribution or partition coefficient.

It is denoted as K_d .

Its formula is

$$K_d = \frac{[S]_{org}}{[S]_{aq}} = \frac{C_1}{C_2}$$

Where $[]$ denotes concentrations of the distributing solute species, S and the value of K_d is independent of the total solute concentration.

Distribution ratio

It is the ratio of the concentrations of all the species of the solute in each phase.

It is denoted by D .

Example:

In the example of benzoic acid (HBz) distribution ratio is given by

$$D = \frac{[HBz]_o}{[HBz]_a + [Bz^-]_a}$$

However, part of benzoic acid in the aqueous

22

Example: The extraction of benzoic acid layer will exist as Bz^- .

from an aqueous solution. In the example of benzoic acid (HBz), the distribution coefficient is given by

$$K_d = \frac{[HBz]_e}{[HBz]_a}$$

Where "e" represents the ether solvent & "a" represents the aqueous solvent.

(X)

In FES low temperature flame is used for alkali & alkaline earth metals:

Low temperature flame is used for alkali and alkaline earth metals in FES because alkali and alkaline earth metals are very reactive metals.

If these metals are exposed to even the slightest heat or flame they ignite immediately. This is because they have ^{1 or 2} free active electron in their outer most shell which they can lose easily. Secondly they have nascent hydrogen which is

very active hydrogen. Due to this hydrogen these metals become active and react very easily on a low flame.